Gehad Abdelghany Demerdash

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CAREER OBJECTIVE

Machine Learning Intern and aspiring **Computer Vision / AI Engineer** leveraging hands-on expertise in PyTorch, OpenCV, and FastAPI to build high-accuracy computer vision models, scalable inference APIs, and end-to-end deployment pipelines for engineering teams.

EDUCATION:

Bachelor of Science in Computer Science and Artificial Intelligence–Capital University(Helwan)

Major: Artificial Intelligence

Grade: very good (Gpa:3.2) *Expected Graduation: 2027*

TECHNICAL SKILLS:

- **Programming Languages:** Python, C, Java, JavaScript, Dart, SQL
- **AI / ML & Computer Vision:** PyTorch, OpenCV, Scikit-learn, SciPy, MobileNetV2, Feature Extraction (GLCM, Hu Moments), Data Augmentation
- **Deployment & Environments:** FastAPI, Streamlit, Docker, Docker Compose, Linux (Ubuntu), WSL 2, Git & GitHub, Firebase
- **Robotics & Simulation:** Webots (e-puck robot controllers, sensor integration, autonomous navigation)
- **Core Competencies:** Problem-Solving, Agile Collaboration, Cross-Functional Communication, Analytical Thinking

KEY PROJECTS

Medical Image Analysis & Feature Extraction Platform | *Python, OpenCV, SciPy, Streamlit, Scikit-learn*

- Developed an interactive medical diagnosis web application in Streamlit for Region-of-Interest (ROI) segmentation and tissue classification.
- Extracted quantitative texture, shape, and statistical descriptors utilizing GLCM and Hu Moments via OpenCV and SciPy.

- Trained and benchmarked a Random Forest classifier, achieving high diagnostic accuracy on segmented biomedical image datasets.

Real-Time Face Mask Detection & API Service | *PyTorch, MobileNetV2, OpenCV, FastAPI, Docker*

- Engineered a lightweight CNN classification pipeline using MobileNetV2 with custom data augmentation and Gaussian noise robustness layers.
- Containerized and deployed inference pipelines into an asynchronous REST API via FastAPI and Docker for reproducible deployment.

Autonomous e-Puck Robotics Simulation | *Python, Webots*

- Designed obstacle-avoidance and line-following navigation controllers in Python for differential-drive e-puck robots inside Webots.
- Integrated real-time proximity sensor feeds to execute robust path-correction and obstacle-handling logic.

TRAINING & CERTIFICATIONS

- **Digital Egypt Pioneers Initiative (DEPI) – AI & Data Science Trainee**
- Completed an intensive, hands-on training program focused on Machine Learning, Deep Learning, and data processing.
- Developed and trained predictive machine learning models using Python, Scikit-Learn, Pandas, and NumPy to solve real-world problems.
- Gained practical experience in freelancing fundamentals, project scoping, and client communication.

LANGUAGES

- **Arabic:** Native
- **English:** Professional Working Proficiency